Elenchus: review edition

Prepared 25 October 2026 for review on 26 October.

The recommended reading set is **Holistic Accuracy and the Lockean Thesis**, **When Acting on Knowledge Makes Free Evidence Harmful**, and **Flattening and Counterfactual Independence**. This is a selection for serious review, not a claim that three original contributions have been established. The first two have the clearest checkable results. The third presents a focused but still disputed interpretation of ordinary counterfactuals.

This edition replaces the older reading packet as the starting point. It incorporates the latest Claude drafts, the earlier review record, and a fresh correction pass. Four manuscripts were substantively revised. The other nine developed tracks and eight exploratory drafts are preserved in the accompanying source archive. Nothing has been submitted to the competition.

Read in this order

1. **Holistic Accuracy and the Lockean Thesis — A5, revision 2.** Start with §3 and then read §§2, 4 and 6. The central example is small enough to check by hand. It challenges the inference from order-independent, truth-directed accuracy maximization to a fixed belief threshold. This is the strongest candidate in this review edition.
2. **When Acting on Knowledge Makes Free Evidence Harmful — A6, revision 2.** Start with the second-weighing example in §3 and the two boundary statements in §4. The proposed contribution is a precise condition for safety across all decision problems, not the familiar observation that treating defeasible knowledge as certainty can cause trouble.
3. **Flattening and Counterfactual Independence — E3, revision 4.** Read the Reunion/Avoidance pair in §3 before the formal material. The revised §5.1 is essential: the linguistic case remains conditional on a stable scenario-relative reading. This is the most vulnerable of the three to a substantive objection rather than an algebraic error.

The combined reading PDF contains this guide followed by those three complete essays. Separate PDFs are supplied for all three and for the causation reserve. The source archive includes editable Markdown, the assessment, methodology records, revision differences, and calculation outputs.

Assessment of the three

A5: a specific counterexample, not the first non-Lockean accuracy theory

What survives. A fixed score over all belief states adds a premium for an error-free record. At credences (0.70, 0.15, 0.15), its unique optimum believes the first world; at (0.75, 0.01, 0.24), its unique optimum does not. The score, option space and proposition are unchanged. The expected accuracies and runner-up margins were reproduced exactly.

The general positive-premium result has a proof, and its sampled parameter cases passed again.

Why it could matter. If the target's minimal premises permit this score, accuracy maximization does not by itself settle the Lockean form of the relation between credence and belief. The dispute must concern further restrictions on epistemic value or rational choice.

Closest predecessors. Dorst's published §4.2 and proof of Proposition 5 were checked. Wang's full 2025 article was recovered; it already treats general, potentially nonadditive utility on deductively cogent belief states. Rothschild's scoring approach and Holguín's nonmonotonicity argument are also acknowledged. The proposed increment is the unrestricted-option counterexample to this particular derivation and the characterization of the record score. Broad claims to have discovered non-Lockean accuracy maximization were removed.

Strongest objection. Sequence-Irrelevance might be intended to require more than order-independent assessment of final states. A defender could demand optimal local choice at every stage, or impose a restriction on which holistic scores count as accuracy. The revision states that interpretive issue and its explicit permission bridge. A small example with correct arithmetic does not settle the interpretation.

Changes. Removed the assertion that every valid route to Lockeanism requires separability. ME plus BI is now presented as one sufficient repair. Corrected the unqualified claim that record-score belief is always consistent. Removed the separate split-invariance theorem and its incompletely established minimality claim. Added the representation-dependence objection. Replaced the abstract-only Wang comparison and old Dorst pagination.

Verdict. Advance for external philosophical review. The restricted claim is promising; the assertion that the target's stated premises are all met requires especially close scrutiny. No priority certification.

A6: an exact boundary with important domain restrictions

What survives. In finite statist, comparable K-normality models, a given question is safe for every decision problem exactly when it has no relative defeat. With the specified questions available, every question is safe exactly when the current ken is gapped. These are different quantifiers. The construction of a three-act counterexample to the stated conditional expected-utility repair also survives the finite checks.

Why it could matter. This gives a precise constraint on a proposed way of making knowledge guide numerical choice. It separates cases where the decision rule can guarantee the value of free checking from cases where some decision problem defeats that guarantee.

Closest predecessors. Goodman and Salow already supply the normality framework and defeat examples. Salow already connects dogmatism with Preservation. Hong's full 2026 paper was recovered and already challenges action guidance using normality-based knowledge. The revised contribution claim is the finite safety boundary and the specified

repair's counterexample, not a first discovery that normality can make action guidance problematic.

Strongest objection. The universal boundary ranges over a generous class of questions and stakes. It does not establish irrational avoidance for every ordinary decision problem or refute every knowledge-first decision theory. The continuous measurement example needs a probability model and a positive-measure argument; a finite-state singleton proof cannot do that work.

Changes. Corrected the empty-question value from "always zero" to "nonnegative," with an explicit two-state counterexample to equality. Corrected the existential quantifier over unsafe questions. Restricted the ten-to-one claim to normal situations. Added a positive-measure argument for Gaussian measurement and identified its conditional prior assumption. Removed an unsupported assertion about generic absence of payoff ties and narrowed the encroachment conclusion. Updated the Hong comparison.

Verdict. Advance alongside A5. Its formal boundary is more secure than its broad philosophical rhetoric was. The revised domain and question-class restrictions are part of the result, not qualifications to omit from a later abstract.

E3: a discriminating case whose interpretation remains unsettled

What survives. The explicit Reunion and Avoidance models agree on all first-degree conditionals at actuality while giving different verdicts to the relevant nested conditional. The controls and finite conservativity construction were rerun successfully. Their correctness does not establish that ordinary speakers interpret the sentences that way.

Why it could matter. A stable scenario-relative reading would challenge the natural-language motivation for Flattening while leaving Dorr and Mandelkern's mathematical characterization intact. The example asks what an embedded supposition inherits from the scenario reached by the outer supposition.

Closest predecessor. Dorr and Mandelkern already know that similarity semantics permits formal countermodels and discuss one themselves. Their discussion in the repository's source extract was checked. The potentially new contribution is the independence/avoidance contrast and its diagnostic controls, not the mere existence of a model where Flattening fails. The external O3 manuscript is another treatment of the same project, not independent novelty evidence.

Strongest objection. A temporary context shift could produce the apparent contrast while preserving Flattening for a fixed operator. Different background stories can shift interpretation without changing sentence wording. The draft's earlier attempt to rule this out was unsuccessful. Locality in Buttons likewise needs a counterfactual selection premise; relativity alone does not fix it.

Changes. Rewrote the abstract, thesis, context-shift reply and conclusion. Removed the assertion that no natural-language interpretation could save Flattening. Stated the local-change premise explicitly and restricted the conservativity observation to the finite models actually used. The paper now offers a testable challenge, not a completed empirical refutation.

Verdict. Provisional third reading candidate. Do not treat its third-place inclusion as evidence that its linguistic claim has been independently confirmed. E1 or E2 remains an alternative if this dispute proves fatal to the proposed contribution.

Causation reserve: A4 revision 2

The frozen-witness lemma previously assumed that an endogenous member of a conjunctive cause had unchanged parents. Another member of the conjunction can be its ancestor, so that inference fails. The revised proof uses acyclicity and AC3 minimality: an equation-conforming witness is already generated by intervening on the root members, making any extra endogenous candidate redundant.

The other main correction concerns scope. Four specified uniform default codings are not every possible uniform coding, and a local componentwise filter is not the whole Halpern–Hitchcock framework. The text now distinguishes those restrictions. The locust/switch contrast is defended conditionally rather than described as indisputable.

The existing case program ran again. The repaired lemma is also supported by its written argument; the finite program is not a proof for all causal models. The manuscript remains a reserve because the philosophical objection can be answered by enriching the representation, rejecting a verdict, or leaving the restricted ranking class. Halpern's full 2016 book remains an incomplete comparison. Its importance does not automatically establish the novelty of this restricted impossibility result.

Inventory and dispositions

The project contains **13 developed tracks, approximately 10 distinct arguments, and eight exploratory drafts**. Revisions, model cards, checks and reading packets are not additional essays. "OA" below distinguishes external OpenAI manuscripts from Claude manuscripts with the same A-code.

Track	Current review source	Disposition
Claude A5	Holistic Accuracy, v2 in this edition	First reading candidate
Claude A6	Free Evidence, v2 in this edition	Second reading candidate
E3	Flattening, v4 in this edition	Provisional third candidate
Claude A4	Normality and Actual Causation, v2 here	Corrected reserve
E1	Gettier Knowledge across Independent Sources, v3	Reserve; Gettier diagnosis and increment beyond the target's own model remain decisive

Track	Current review source	Disposition
E2	Ignorable Learning, external v5	Reserve; newest version addresses Gallow and Reflection scope, but needs a fresh full referee decision
O1	Independent Sources and Correlations	Alternate E1 treatment, not a separate discovery
O2	Bayes and the Benchmark	Alternate E2 treatment, not a separate discovery
O3	Independent Interventions	Alternate E3 treatment, not a separate discovery
O4	Evidence and Description, v2	Do not advance as a refutation of a claim its target does not make
O5	How Regret Selects a Learning Standard, v3	Reserve; classification is narrower than generic minimax/score-dependence claims
OA-A4	Finite relay criterion, v2	Research note; no established full apt counterexample
OA-A6	Almost-sure recovery and cost, v0.2	Research note; adequate epistemic realization still missing

The last nine tracks are archived as received, not silently certified by this pass. Claude's earlier negative reviews remain relevant; later revisions are not automatically cleared by answering some objections. The previous E1/E2/E3 preparation set was provisional as well.

The eight exploratory ideas remain: usable revocation; models and accountability; preserving correction; inherited authority; translating disagreement; forgotten obligations; reopening inquiry; and contesting objectives. Their existing audit demotes broad novelty claims and recommends merging preserving correction with reopening inquiry. C045, on claimant assistance after memory loss, remains a research prospect rather than a finished contest entry. The source archive preserves that audit and its outstanding text requests. Further expansion of this pool was not necessary to finish a useful review edition.

Verification and its limits

Fresh checks reproduced A5's exact optima, nine robustness cases, and the core characterization on 887 sampled instances. A6's program examined 64,004 comparable situation–question pairs through four states: all 3,306 relative-defeat cases yielded the constructed negative-value examples, and no sampled KDT failure occurred without relative defeat. Gaussian integration reproduced approximately -0.0499, -0.1202 and

−0.2739 for the displayed 10:1, 20:1 and 100:1 bets. The empty-question correction gives +0.5 in its explicit example. E3's case, control, conservativity and crashing-fragment checks passed. A4's case program completed and is retained.

These are targeted reruns of the existing calculations, not a new software project. The optional Z3 dependency was unavailable in this runtime. E3's historical solver claim was not rerun; its mathematical argument and dependency-free checks were inspected instead. Sampling does not prove general claims, and computation does not prove originality or philosophical significance. Detailed outputs and limitations are in VALIDATION.md.

Source access and review standard

Dorst's published article, Wang's full article, and Hong's full article were accessible in this pass. Those earlier access gaps are closed for the comparisons reported here. Dorr and Mandelkern's relevant discussion was checked through the repository's text extract. No source was treated as read merely because a search result named it.

One additional comparison located in this pass remains inaccessible: **Ginger Schultheis, "Reasonable Inferences for Counterfactuals"** (manuscript, PhilArchive SCHRIF-3). The indexed description concerns reasonable counterfactual inference; its appearance under a Flattening search is not evidence that it contains this result. A PDF would permit the comparison. It is recorded as an unresolved literature lead, not an alleged refutation. The older reserve-track requests, including the published Darwall material and Halpern's book, remain in SOURCE-REQUESTS.md. None prevents reading the completed packet tomorrow.

The assessment applies four questions: what exactly follows; which premise could reasonably be rejected; what the nearest predecessor already establishes; and what philosophical belief would change if the result were accepted. These are synthetic adversarial tests by the revising assistant. No philosopher, contest judge, future model or independent external reviewer has endorsed this edition.

This is the stopping point for this review edition. The unresolved questions are documented rather than hidden behind extra word count or a simulated score. The next substantive judgment belongs to an independent review of the narrowed claims.

Holistic Accuracy and the Lockean Thesis

Abstract. Does maximizing expected accuracy force a Lockean threshold rule for belief even when accuracy is holistic? Dorst argues that it does. I identify a gap in the derivation and give a counterexample on the unrestricted space of belief states. A score that adds a premium for having no false beliefs is strictly truth-directed and makes the rational assessment of final states independent of their order of formation. Yet, for every positive premium, it permits belief in a proposition at one credence and requires withholding it at a higher credence. The optimum combines a selected core hypothesis with a probability threshold. I also characterize a pair of sufficient repairs to the derivation: together they recover separability on states without complementary pairs. This does not establish that separability is necessary for every Lockean representation. It establishes that truth-directed accuracy and order independence alone do not settle the form of the bridge from credence to belief.

1. Introduction

Accuracy-first epistemology holds that rational doxastic states are those that best promote accuracy by the agent's own lights. With a separable score for outright belief, positive for true beliefs and negative for false ones, it yields a Lockean thesis: believe a proposition just in case your credence in it exceeds a threshold. The separable version of this result is due to Easwaran (2016; see Dorst 2019, §3). Dorst (2019) extended it in two directions. Letting the values of true and false belief vary with proposition and context yields variable thresholds (his Corollary 2). And, he argued, dropping Separability, the assumption that a belief state's accuracy is the sum of the accuracies of its members, changes nothing: given a further premise, Sequence-Irrelevance, Lockeanism follows anyway (his Proposition 5). The resulting claim is that the Lockean form survives a substantial relaxation of separability.[1]

The broad robustness claim needs qualification. The counterexample below preserves unrestricted choice, truth-directed value, and order-independent assessment of final states. It nevertheless violates a fixed, proposition-specific threshold rule. Theorem 2 establishes that violation for every positive value of a premium for an error-free record. Theorem 3 characterizes the resulting optima. Theorem 1 diagnoses the derivation by identifying a pair of sufficient additional assumptions that restore a separable score. It does not prove that every successful derivation of Lockeanism must use them.

The thesis does not say that Lockeanism or separability is false. Nor does it say that holistic accuracy guarantees consistency or deductive closure: the characterization below states exactly when those properties hold. The question is what follows from the advertised minimal premises.

The closest prior work makes the novelty claim narrow. Rothschild (2023) develops additive scoring representations of Lockean belief and explicitly invites investigation of weaker additivity assumptions. Wang (2025, §§2 and 6) studies general expected-utility rules on deductively cogent belief states; his framework permits nonadditive utility and

[1] The published text of Dorst (2019), especially pp. 192–195 and 206–207, was checked for this revision. Earlier draft page references used the accepted manuscript.

derives convexity constraints. Non-Lockean expected-utility belief is therefore already established. The present contribution sought is an unrestricted-option counterexample directed at Dorst's §4.2, together with a characterization of one holistic score. It does not obtain its result by excluding inconsistent or nonclosed states. Holguín (2022, §5) independently argues that rational thinking need not be monotone in credence. Here that behavior is obtained within an accuracy-maximizing model.

The stakes extend beyond formal epistemology. If accuracy alone settled the bridge between credence and belief, the Lockean would win that debate by default. If, as I argue, the shape of the bridge depends on how we value combinations of true beliefs, then the debate over Lockean, closed and guess-like belief becomes a debate about epistemic value, and accuracy-first epistemology is a framework for conducting it rather than a verdict.

2. Dorst's derivation

Let W be a finite set of worlds, let propositions be subsets of W, and let a belief state B be any set of propositions. Accuracy First leaves inconsistent, nonclosed and empty belief states among the alternatives (Dorst 2019, §3). An accuracy measure assigns a real number $A(B, w)$ to each state at each world. Given a probability function C on W, the expected accuracy is $EA_C(B) = \sum_w C(w)A(B, w)$, and Maximize Expected Accuracy says that B is rational only if no state has higher expected accuracy.

Under Separability with proposition-dependent values, $A(B, w) = \sum_{p \in B} v_p(w)$, where $v_p(w) = T_p > 0$ if $w \in p$ and $v_p(w) = F_p < 0$ otherwise. Then $EA_C(B) = \sum_{p \in B} e_p$, where

$$e_p = C(p)T_p + (1 - C(p))F_p.$$

Every optimum includes p when $e_p > 0$ and excludes it when $e_p < 0$; inclusion is optional at equality. The threshold is $\tau_p = -F_p/(T_p - F_p)$ (Corollary 2). Dorst also imposes Conservativeness: in the uniform case $|T| < |F|$, which makes $\tau > 1/2$.

Dorst's §4.2 allows the value of a whole belief state to depend on its other beliefs. He considers adding a proposition to a background state, first when it is true and then when it is false. The argument assigns a positive true-belief increment and a negative false-belief increment, uses them to derive a threshold for the final addition in a sequence, and invokes order independence to extend the threshold to belief generally (2019, pp. 192–195, 206–207).

There are two distinct issues here. A true addition can improve a score at every world where it is true without improving it by the same amount at all those worlds. And invariance under rearranging a sequence does not by itself make the value of an addition independent of which other propositions the final state contains. Section 4 separates these claims.

I read Sequence-Irrelevance as a constraint on rational assessments: holding the other factors fixed, changing the order in which a final state was formed does not alter its rational status. A single objective over whole states satisfies that constraint. It need not make greedy, locally optimal additions lead to a globally optimal state in every order. Requiring that stronger property would constrain the optimization procedure as well as

the rationality of its outcome. The counterexample addresses the former, stated order-independence principle; a defender may instead adopt the stronger requirement, but must defend it as an additional restriction on holism.

3. The counterexample

Define the record measure with premium $\beta > 0$:

$$A_\beta(B, w) = \sum_{p \in B} v_p(w) + \beta \cdot [\, w \in \bigcap B \,],$$

where $\bigcap \emptyset = W$ and $[\cdot]$ is 1 if the bracketed condition holds and 0 otherwise. A belief state earns the separable score of its members, plus β at exactly those worlds where everything it contains is true. The premium is the value of an error-free record. One rationale is inferential: a belief state whose members are all true is one from which any amount of multi-premise deduction yields only truths. A body of belief can thus be valuable as a whole, which is the holist's thought. Dorst motivates separability by an exam scored question by question (2019, §3). Many exams also reward a perfect paper, and the record measure scores belief that way. The measure is purely alethic: $A_\beta(B, w)$ depends only on B and on which of its members are true at w.

It meets the stated conditions of Dorst's §4.2 on the final-state reading just specified. Adding a true p raises A_β by exactly T_p: if every member of B_0 was true at w, all still are. Adding a false p lowers it by $|F_p|$, or by $|F_p| + \beta$ if B_0 was error-free at w. So truth-directedness holds. Conservativeness can be imposed on the v_p. The measure does not depend on the current state, so Maximize Expected Accuracy has a single, sequence-independent verdict and Sequence-Irrelevance holds. Every belief state is an option, so Accuracy First holds. I make the following permission bridge explicit:

(Permission) If a belief state uniquely maximizes expected accuracy, it is rationally permitted.

This is an explicit bridge from optimization to permission. A merely necessary accuracy constraint, combined with unspecified further prohibitions, would not license that inference. The counterexample concerns an accuracy-first theory that permits its unique optimum.

The case. Let $W = \{w_1, w_2, w_3\}$, $T_p = 1$ and $F_p = -2$ for every p (so $\tau = 2/3$), and $\beta = 3$. Write $\mathbf{1} = \{w_1\}$. At $C = (0.70, 0.15, 0.15)$ the unique optimum among all 2^8 belief states is $\{\mathbf{1}, \{w_1, w_2\}, \{w_1, w_3\}, W\}$, with expected accuracy 4.30 against a runner-up of 4.20. At $C' = (0.75, 0.01, 0.24)$ the unique optimum is $\{\{w_1, w_3\}, W\}$, with 4.94 against 4.75. Hand computation agrees with exhaustive enumeration in exact arithmetic.[2] So $\mathbf{1}$ is believed at credence 0.70 and not at credence 0.75. By Permission both optima are rational. Proposition 5 would then require $\tau_{\mathbf{1}} \leq 0.70$ and $\tau_{\mathbf{1}} \geq 0.75$, which is impossible.

This is not a knife-edge.

[2] All 256 states over the eight propositions of a three-world space (including $\emptyset$ and W) were enumerated with rational arithmetic. A scan of credences on a grid of mesh $1/40$ found monotonicity violations for all six contingent propositions.

Theorem 2. Let $T > 0 > F$ with $T < -F$, let $\beta > 0$, let $D = T - F$ and $\tau = -F/D$, and put

$$c_0 = \frac{\beta - T - 2F}{2D + \beta}, \qquad c_1 = \frac{\beta - 2F}{2D + \beta}.$$

Then $\max(c_0, \tau) < c_1 < 1$. For all $c < c'$ in the interval $(\max(c_0, \tau), c_1)$, **1** belongs to the unique optimum at $(c, \frac{1-c}{2}, \frac{1-c}{2})$ and to no optimum at $(c', \varepsilon, 1 - c' - \varepsilon)$, for all sufficiently small $\varepsilon \geq 0$.

The proof is in the Appendix; it was also checked by enumeration for premiums from 0.01 to 1000. The mechanism is visible in the case. At C the residual 0.30 is spread over two worlds, and the hypothesis **1** is the best "core": believing everything it entails earns three positive separable terms and a premium on 0.70. At C' almost all the residual sits on w_3, so the disjunction $\{w_1, w_3\}$ is nearly certain. Its premium on 0.99 outweighs what the more specific hypothesis adds, so the agent hedges. As β shrinks the interval narrows (its length above τ is $\beta T/(D(2D + \beta))$), but it never closes.

The result instantiates Holguín's Non-Monotonicity: a lower credence in a proposition can permit thinking it while a higher credence does not (2022, §5). His argument against Lockeanism used this principle. The record measure shows that accuracy-first epistemology without Separability is not bound by monotonicity either.

4. Diagnosis

Where does the derivation fail? There are two gaps.

The quantifier gap. Truth-directedness gives, for each world w, some positive increment. Dorst's passage on p. 193 then names one number $T_p^{B_0}$ as "the" increment. That moves from "for every w there is a t" to "there is a t for every w." Under the record measure the step fails. Adding the false **1** to $\{\{w_1, w_3\}\}$ costs 2 at w_2 but 5 at w_3, since only at w_3 was the prior state error-free. In expectation, the comparison that Proposition 4 makes between B and $B \setminus \{p\}$ becomes

$$EA_C(B) - EA_C(B \setminus \{p\}) = e_p - \beta\, C(\textstyle\bigcap(B \setminus \{p\}) \setminus p).$$

Belief in p is therefore worth adding iff e_p exceeds a term fixed by the credence of the rest of the core outside p. That is a threshold on $C(p)$ that moves with credences in other propositions, not one fixed by values. What the step needs is:

(ME) Marginal extensionality. For every B_0 and $p \notin B_0$, $A(B_0 \cup \{p\}, w) - A(B_0, w)$ depends on w only through whether $w \in p$.

This is not a consequence of A's being well-defined. It is a substantive claim: the marginal value of a belief may depend on the world only through that belief's own truth value.

The content gap. Grant ME. Proposition 4 then yields thresholds $\tau_p^{B^-}$ indexed to B^-, the rest of the belief state. The proof of Proposition 5 generalizes from "any" B^- to all rational states. That step requires $\tau_p^{B^-}$ not to vary with B^-. Sequence-Irrelevance cannot supply this, because B^- is not a sequence; it is the content of the other beliefs, which is exactly what the holist says matters. For any fixed measure, Maximize Expected Accuracy

compares whole states, and its verdicts are already sequence-independent. Dorst himself distinguishes holism from dependence on the order of consideration (2019, p. 195). That is why a principle about order cannot exclude the holism of content. A sufficient strengthening is:

(BI) Background-independence. $T_p^{B_0}$ and $F_p^{B_0}$ do not depend on B_0, for states B_0 not containing the complement of p.

Together, these two assumptions recover separability on the stated domain. BI is stronger than simply requiring the ratio that fixes a threshold to be background-independent. The theorem characterizes this sufficient repair, not every possible repair.

Theorem 1. Let A be truth-directed. Consider only states containing no proposition together with its complement.[3] (a) A satisfies ME iff $A(B,w) = A(\emptyset,w) + \kappa(B) + \sum_{p\in B} J_p\,[w \in p]$ for some world-independent κ and constants $J_p > 0$, the sum running over the contingent $p \in B$. (b) A satisfies ME and BI iff κ is additive, that is, iff $A(B,w) = A(\emptyset,w) + \sum_{p\in B} v_p(w)$ for some proposition-dependent values v_p.

Since $A(\emptyset,w)$ is the same for every state, it never affects which states are optimal. Part (b) therefore says that adding ME and BI returns us to the separable measures covered by Dorst's Corollary 2. It does not classify every measure whose optima happen to be Lockean. The proof (Appendix) turns on one observation. Adding p and q in either order gives the same total, so $\Delta_p^B - \Delta_p^{B\cup\{q\}} = \Delta_q^B - \Delta_q^{B\cup\{p\}}$, where Δ_r^B is the increment of adding r to B. Under ME the left side is affine in $[w \in p]$ and the right side affine in $[w \in q]$. Unless q is p's complement or trivial, both sides must be constant. A linear-algebra computation over all 256 states on three worlds confirms (a), and it confirms that the complement proviso cannot be dropped.[4]

Part (a) shows what survives of holism under ME: a world-independent term $\kappa(B)$, a bonus or malus for a state's shape regardless of how the world is. A nonadditive κ can make Proposition 5 fail. For example, a small bonus for deductively closed states, small enough to preserve truth-directedness, shifts the effective threshold of p according to whether adding p completes or breaks closure.[5] Dorst can reject such a κ as non-alethic. Rejecting such terms would require a further account of which forms of holistic value qualify as accuracy.

Theorem 1 concerns the derivation, not its conclusion. Some non-separable measures do have Lockean optima. A separable measure lowered on states that are never optimal is an example. So Separability is not necessary for Lockean outputs. What the combined results show is narrower: the proposed inference is not secured by well-definedness and order independence alone. The record measure supplies a truth-directed counterexample, while ME plus BI supplies one transparent repair.

[3] The proviso is needed, since there are ME measures in which the marginal value of p changes when its complement is believed. It is harmless here. Under Variable Conservativeness $\tau_p + \tau_{\neg p} > 1$, so such states are never optimal for separable measures (Dorst's Proposition 3), nor, by Theorem 3 below, for the record measure.

[4] The ME constraints form a homogeneous linear system on the 768 values $A(B,w)$. Each vector of a basis of its solution space was tested against the representation in (a), with numerical deviation below 10^{-14}.

[5] With $T = 1$, $F = -2$ and a bonus of $1/2$ for closed states, the measure satisfies ME and truth-directedness. On three worlds, $\{w_1\}$ is believed at $(0.575, 0.2, 0.225)$ but not at $(0.65, 0, 0.35)$.

5. What holism delivers instead

The record measure does not leave belief unstructured. Its optima have an exact description. Write $\bigcap B$ for the core of B, and for each proposition X let

$$\Phi(X) = \sum_{p \supseteq X} \max(e_p, 0) + \beta\, C(X).$$

Theorem 3. For any finite W and any proposition-dependent values, $\max_B EA_C(B) = \max_X \Phi(X)$. A state B is optimal iff its core maximizes Φ, B contains every $p \supseteq \bigcap B$ with $e_p > 0$, and B contains no p with $e_p < 0$. In particular, when the values are uniform and C is generic, the unique optimum is

$$B^* = \{p \supseteq X^* : C(p) > \tau\},$$

where X^* maximizes Φ, and: (i) B^* is consistent iff $X^* \neq \emptyset$, and it is closed under single-premise entailment; (ii) X^* contains every world more probable than any world it contains; (iii) B^* is deductively closed iff $C(X^*) > \tau$, in which case it consists of exactly the consequences of X^*.

Call this core-relative Lockeanism. The agent selects a working hypothesis X^*, trading its probability (the premium term) against the number of above-threshold propositions it entails. She then believes those above-threshold propositions that the hypothesis entails. Lockean belief is the special case in which the best core is the intersection of all above-threshold propositions. By (ii) the working hypothesis is always a set of most probable worlds, which is the filtering structure guess-based theories posit for answers (Holguín 2022, §10). By (iii) closure is not guaranteed. On four worlds with $C = (0.62, 0.18, 0.10, 0.10)$ and the values of §3, the unique optimum believes $\{w_1, w_2\}$ and $\{w_1, w_3\}$ but not $\{w_1\}$. The theorem was checked against exhaustive enumeration on three and four worlds, with uniform and with randomly varying values.[6]

Three consequences matter for the debate.

First, accuracy-first epistemology is neutral between Lockean and guess-like belief. Separable value gives thresholds. Record value gives a best hypothesis and its probable consequences. Accuracy does not choose between them; a view about the value of an error-free body of belief does.

Second, the record measure captures both forms of non-monotonicity in the literature. In §3 the agent hedges when a disjunction becomes nearly certain. In Holguín's horse race, belief in A is lost when a rival overtakes it: told that A, B and C have chances 0.40, 0.35 and 0.25, one may think A wins; told 0.45, 0.50 and 0.05, one should think B wins and not A (2022, §5). A single context reproduces both verdicts. Let values depend only on how many horses a proposition mentions: $(T, F) = (1, -0.66)$ for one horse and $(0.65, -1.01)$ for two, which satisfies Dorst's Variable Conservativeness, and set $\beta = 0.19$. The unique optimum at the first credences is the set of consequences of "A wins". At the second it is the set of consequences of "B wins". The runner-up there is the inconsistent state that be-

[6] In the revision rerun, 887 random instances with integer credences gave identical predicted and computed optimal sets (799 on three worlds and 88 on four). These finite checks supplement the proof.

lieves both A and B, which is exactly what a Lockean with these thresholds (about 0.40 for one horse) would believe.[7] So the premium does what Holguín says sophisticated Lockeanism cannot: it keeps thresholds low enough for extreme weakness while ruling out the incompatible beliefs those thresholds would license. The margins are small, though, and a random search suggests that values doing this occupy a narrow band. The result shows that non-monotonicity is available to accuracy-firsters. It does not show that the record measure is the right theory of thinking.

Third, Dorst's reduction of belief to credence (§5.1) survives in a weakened form. Under the record measure, belief can still be "read off" credence and contextual values, since B^* is a function of both. What fails is the Lockean form of the function. The reduction is thus independent of Lockeanism. A Lockean needs some further restriction on the admissible scores or rational states. Separability is one sufficient restriction; it is not shown here to be necessary.

6. Objections

The premium is not accuracy; it double-counts the conjunction. The premium is alethic, since it depends only on which believed propositions are true. It is not a disguised belief in the core: in the four-world case it accrues on $\{w_1\}$, which is not believed. And it respects strict truth-directedness, the only constraint Dorst places on holistic measures. The objector may insist that genuine accuracy is local, so that a belief's contribution depends only on its own truth. That insistence is ME. Coupled with BI it yields separability, but ME alone leaves the additional term identified in Theorem 1(a). To exclude the premium on those grounds is to deny the holism that §4.2 set out to accommodate. The reply leaves one thing open: someone may find the premium unmotivated as a value. Then the dispute is about the value of an error-free record. That is my thesis.

Dorst meant only belief-dependence: values that vary with what else is believed, not with whether it is true. The text resists this reading by claiming that well-definedness alone supports the step (2019, p. 193). Grant the reading anyway. ME then becomes an explicit premise, and the content gap remains. Under ME, Theorem 1(a) leaves a world-independent κ. Nonadditive choices of κ can defeat a fixed threshold; additive ones recover separability. This alternative reading narrows the admissible holism but does not, without further argument, establish the advertised conclusion.

Sequence-Irrelevance should be read strongly, as requiring background-independent thresholds. This is a substantive alternative reading, but it goes beyond invariance under reordering an exam. Background-independent thresholds need not be identified with background-independent marginal values. Theorem 1 concerns the latter, stronger condition. A defender who imposes threshold independence directly owes an argument for why an accuracy holist must accept it; appealing only to order independence leaves the issue unsettled.

Maximize Expected Accuracy gives only a necessary condition; perhaps neither optimum is rational. Then rationality would be unsatisfiable at C and C'. Accuracy First would

[7] Exhaustive enumeration over all 256 belief states, in exact arithmetic, with $T_W = 1$ (the value of believing W does not affect which states are optimal) and $\beta = 19/100$. Expected accuracies: 1.384 against 1.380 at the first credences, and 1.832 against 1.824 at the second.

then have yielded a norm that no state can meet, which no accuracy-firster accepts. Permission is the minimal premise needed for the framework to issue verdicts at all.

The premium is representation-dependent. The value of a record may change when the proposition space is refined: the additive part counts more propositions, while the premium is paid once. This is a serious objection to treating the measure as a complete account of epistemic value. The counterexample fixes a finite proposition space, the same unrestricted space used in the target derivation, and compares two credences on it without changing the questions or their values. Its nonmonotonicity therefore does not arise from switching representations between the two cases. But nothing proved here establishes invariance across different descriptions of the same inquiry. A positive theory based on record value would need to say which questions count and how their weights survive refinement. The limited negative result does not depend on having solved that further problem: the proposed derivation must either exclude the score by a stated premise or accommodate its verdicts. Calling that exclusion a representation-invariance requirement would make the substantive restriction visible.

The result is negative only; Dorst's picture survives. In part, yes. Belief remains a function of credence and contextual values, and Dorst's "best guess" gloss fits core-relative Lockeanism at least as well as Lockeanism. What does not survive is the claim that accuracy forces the Lockean form. That matters to Holguín's argument, which targets the Lockean form. Non-monotonicity tells against sophisticated Lockeanism, but not against accuracy-first epistemology as such.

7. Conclusion

The record measure gives a precise obstruction to the claimed inevitability of Lockean belief. It evaluates every belief state, rewards true additions, penalizes false additions, and judges final states without regard to formation order. With the explicit permission bridge, it nevertheless licenses a nonmonotone relationship between credence and belief for every positive premium. Its optimum has an exact core-and-threshold description; consistency and closure require the additional conditions stated in Theorem 3.

The diagnosis is deliberately limited. ME plus BI restores separability on the specified domain, but separability has not been proved necessary for every Lockean output. Nor has the record premium been established as the uniquely correct measure of accuracy. What changes is the burden of argument: a defence of Lockeanism must justify a restriction that excludes this kind of holistic value, rather than taking order-independent accuracy maximization to have excluded it already.

Appendix: Proofs and checks

Proof of Theorem 1. Write $\Delta_p^B(w) = A(B\cup\{p\}, w) - A(B, w)$ for $p \notin B$. Under ME, $\Delta_p^B = F_p^B + J_p^B[w \in p]$ for contingent p, and Δ_p^B is constant for $p \in \{\emptyset, W\}$; truth-directedness gives $F_p^B < 0 < F_p^B + J_p^B$. (i) Let $p, q \notin B$ be distinct, with q not the complement of p. Computing $A(B \cup \{p, q\}, \cdot) - A(B, \cdot)$ in both orders gives $\Delta_p^B - \Delta_p^{B\cup\{q\}} = \Delta_q^B - \Delta_q^{B\cup\{p\}}$. The left side is $\alpha + (J_p^B - J_p^{B\cup\{q\}})[w \in p]$ and the right side is $\gamma + \delta[w \in q]$, or a constant if

q is trivial. If p, q are contingent, distinct and not complementary, then $1, [w \in p], [w \in q]$ are linearly independent, so $J_p^{B \cup \{q\}} = J_p^B$. If q is trivial, the same conclusion follows. (ii) Building any B that contains neither p nor its complement from $\emptyset$ one member at a time, (i) gives $J_p^B = J_p^\emptyset =: J_p$. (iii) For B without complementary pairs, add its members in any order. Each increment is $F_p^{B_i} + J_p[w \in p]$, so $A(B, w) - A(\emptyset, w) = \kappa(B) + \sum_{p \in B} J_p[w \in p]$, where $\kappa(B)$ is the sum of the F-terms (and the constant increments). It is world-independent, and it is order-independent because the total is. The converse of (a) is immediate. For (b), BI says $F_p^B = \kappa(B \cup \{p\}) - \kappa(B)$ is independent of B, which holds iff $\kappa(B) = \kappa(\emptyset) + \sum_{p \in B} f_p$; then $v_p = f_p + J_p[w \in p]$. Conversely, separable measures satisfy ME and BI trivially. □

Proof of Theorem 3. $EA_C(B) = \sum_{p \in B} e_p + \beta C(\bigcap B)$. Every member of B includes $\bigcap B$, so $EA_C(B) \leq \Phi(\bigcap B)$, with equality iff B contains every $p \supseteq \bigcap B$ with $e_p > 0$ and no p with $e_p < 0$. Conversely, given X, let $B_X = \{p \supseteq X : e_p > 0\}$ and $X' = \bigcap B_X \supseteq X$. The members of B_X are exactly the positive-e propositions including X', so $EA_C(B_X) = \Phi(X') \geq \Phi(X)$, because $C(X') \geq C(X)$. Hence $\max EA = \max \Phi$, and the optimality condition follows. For uniform values, upward closure holds because $p \subseteq q$ implies $C(q) \geq C(p)$. For (ii): if $u \in X^*$, $v \notin X^*$ and $C(v) > C(u)$, swapping u for v maps the supersets of X^* bijectively onto those of the new core without lowering any credence, and it raises $C(X^*)$; that contradicts maximality. For (iii): B^* is upward closed with $\bigcap B^* = X^*$, so it is closed under conjunction iff $X^* \in B^*$, iff $C(X^*) > \tau$. Consistency holds iff $\bigcap B^* \neq \emptyset$. □

Proof of Theorem 2. Write $e_p = C(p)D + F$; note $\tau > 1/2$. At $C_c = (c, \frac{1-c}{2}, \frac{1-c}{2})$ with $c > \tau$, the positive terms are $e_{\mathbf{1}}$, $e_{12} = e_{13}$ and $e_W = T$, and all others are negative (their credences are at most $1 - c < 1/2$). Then $\Phi(\mathbf{1}) - \Phi(W) = e_{\mathbf{1}} + 2e_{12} - \beta(1-c) = c(2D+\beta) + T + 2F - \beta$, which is positive iff $c > c_0$. Also $\Phi(\mathbf{1}) - \Phi(\{w_1, w_3\}) = e_{\mathbf{1}} + e_{12} - \beta\frac{1-c}{2} > \frac{1}{2}(\Phi(\mathbf{1}) - \Phi(W))$, which is positive. Next, $\Phi(\mathbf{1}) - \Phi(\{w_2\}) = e_{\mathbf{1}} + e_{12} + \beta\frac{3c-1}{2} > 0$ and $\Phi(\mathbf{1}) - \Phi(\{w_2, w_3\}) = e_{\mathbf{1}} + 2e_{12} + \beta(2c-1) > 0$; the cases of $\{w_1, w_2\}$ and $\{w_3\}$ are symmetric; and $\Phi(\mathbf{1}) - \Phi(\emptyset) = \beta c > 0$. By Theorem 3 the unique optimum has core $\mathbf{1}$ and contains $\mathbf{1}$. At $C' = (c', 0, 1-c')$ with $\tau < c'$: $e_{\mathbf{1}} = e_{12} = c'D + F > 0$ and $e_{13} = e_W = T$, so $\Phi(\{w_1, w_3\}) - \Phi(\mathbf{1}) = \beta(1-c') - 2e_{\mathbf{1}}$, which is positive iff $c' < c_1$, while $\Phi(\emptyset) = \Phi(\mathbf{1}) - \beta c'$. A state containing $\mathbf{1}$ has core $\mathbf{1}$ or $\emptyset$, so no optimum contains $\mathbf{1}$. These strict inequalities persist for small ε. Finally, $c_1 - c_0 = T/(2D+\beta)$, $c_1 - \tau = \beta T/(D(2D+\beta))$ and $1 - c_1 = 2T/(2D+\beta)$ are all positive. □

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When Acting on Knowledge Makes Free Evidence Harmful: Good's Theorem in Normality Models

Abstract. Knowledge-based decision theory tells an agent to maximize expected utility with probabilities conditioned on what she knows. I examine this rule inside Goodman and Salow's normality theory of inductive knowledge. In finite statist, comparable models, with K-normality and the specified questions available, it gives every free question non-negative value at a situation if and only if the agent's knowledge is gapped: each possibility she leaves open is sufficiently more normal than each evidential possibility she rules out. More generally, any factive state that leaves open whatever rational belief leaves open can be safely conditioned on only if it is gapped. In graded models a factive gapped state must coincide with the evidence. A separate positive-measure argument extends the negative result to their Gaussian measurement model; at normal situations a second weighing has negative value at ten-to-one stakes. An expected-utility form of Salow's Conditional Knowledge Dominance does not help. It fails wherever the unconditional rule fails, and in non-statist models its verdicts turn on how conditional knowledge is read.

1. Introduction

If you know p, any evidence against p is misleading, so it seems you may avoid it. This is the dogmatism puzzle. Salow (2025) offers the most economical recent dissolution. If Preservation holds (discovering something consistent with your beliefs never requires giving one up), you cannot believe both p and that you might meet evidence defeating p, so the puzzle never arises. If Preservation fails, the puzzle needs Weak Knowledge Dominance, which Salow replaces with a principle that looks equivalent but is not:

> Conditional Knowledge Dominance: If you know that B's consequences would be better than A's if they're not equally good, you should choose B over A. (Salow 2025, 33)

In Salow's Suitcase, you do not know that the suitcase weighs under 18 kg on the hypothesis that it weighs over 17.5 kg, since discovering the latter would defeat that knowledge. So the principle does not tell you to avoid checking. Salow's principles are qualitative. He grants that how all-out attitudes guide choices made under uncertainty about outcomes is "a general problem", and he expects that "Any solution to that problem will generalize to this one" (2025, 34).

The most developed solution of that kind is knowledge-based decision theory (KDT): maximize expected utility using probabilities conditioned on what you know, so that known propositions get probability 1. Goldschmidt (2024) axiomatizes a version due to Hawthorne and Stanley (2008), and Hong (2024) explores the view. I ask whether KDT keeps the practical content of Salow's dissolution, namely Good's (1967) theorem that free evidence never has negative expected value, once knowledge is modeled by the most developed formal theory of inductive knowledge, the normality framework of Goodman and Salow (2023, 2025). My thesis is this:

> In finite statist models with the specified questions available, conditioning on a factive belief-containing state makes every question safe only if that state is gapped. Under K-normality and comparability the condition is also sufficient. The proposed conditional expected-utility repair has counterexamples wherever the unconditional rule does.

Three results support the thesis. First, an exact boundary (Theorem 1). With K-normality, statism and comparability, KDT gives a free question non-negative value for every decision problem if and only if no situation compatible with what one knows is one where answering would reopen a possibility one now rules out. Every question has non-negative value if and only if knowledge is gapped. Theorem 2 extends the necessity half to any factive state that leaves open what rational belief leaves open, whichever theory of knowledge one adopts. Second, ubiquity. In Goodman and Salow's own model of Bjorn weighing himself twice, some bet makes KDT price the free second weighing negatively at each situation. At normal situations the displayed bet already does so at ten-to-one stakes (§3). Third, the conditional repair fails (§5). I define the expected-utility analogue of Conditional Knowledge Dominance. It handles the bets that motivate it. But in any situation where KDT disvalues a question, some decision problem makes the conditional rule disvalue it too (Theorem 3). The proof uses a perturbation that removes payoff ties between the selected act now and the later policy. It does not assert that ordinary decision problems generally lack such ties.

The closest prior work is as follows. Salow (2025) shows that Weak Knowledge Dominance licenses dogmatism exactly where Preservation fails, and he calls such failures "relatively rare" (36). I give the expected-utility counterpart, with an exact boundary, and show that in graded models failures of Good arise at every situation with inductive knowledge. I also show that his remedy does not carry over to expected utility. Goodman and Salow (2025, henceforth BRN) show that statism, collapse and comparability validate Preservation and that Preservation fails in Bjorn. I give the decision-theoretic counterpart, together with what follows for decisions from their distinction between conditional and post-discovery belief. Geanakoplos (1989), as reported by Das (2023), gives conditions on information structures that suffice for Good's theorem. My contribution is the bridge from normality kens to those conditions, and the result that no belief-containing state other than a gapped one meets them. Hong (2026, §§1–3) directly criticizes normality-based action guidance through lottery switching, sweeteners and the value of otherwise irrelevant information. Those problems already challenge conditioning on normality knowledge. The proposed increment here is an exact finite safety boundary and a construction defeating the specified conditional expected-utility repair, rather than the general claim that normality creates decision-theoretic difficulties.

In plain terms: someone who treats what she has inductively learned from a measurement as settled when she bets will, in the leading model of measurement, sometimes pay to avoid a free second look. The fault lies in how the theory of knowledge interacts with expected utility, not in the bettor.

The essay does not claim that knowledge lacks a practical role, that Good's theorem is sacrosanct, or anything about Salow's qualitative principles taken alone. The result matters beyond formal epistemology. If inductive knowledge, as the leading theory mod-

els it, cannot set the odds without making free checking harmful, then knowledge-first accounts of practical reason must explain which modeling assumption or decision rule should change. §2 sets out the framework, §3 the case, §4 the theorems, §5 the conditional repair, §6 objections. Proofs are in the appendix.

2. Framework

I follow Goodman and Salow (2023) and BRN. A normality structure has a set S of states, a family $\mathcal{E}$ of possible bodies of evidence, and two relations on situations $\langle s, E\rangle$ (with $s \in E \in \mathcal{E}$): $\succeq$ (at least as normal as), a preorder, and $\gg$ (sufficiently more normal than). $\gg$ implies $\succeq$, is irreflexive, and chains: if $w_1 \succeq w_2 \gg w_3 \succeq w_4$ then $w_1 \gg w_4$. Under statism both relations are evidence-independent, so I treat them as relations on states. Comparability says any two states with the same evidence are $\succeq$-related. Collapse says every strict difference in normality is sufficient. For the theorems S is finite.

Rational belief, given evidence E, rules out exactly the evidential possibilities that something else in E is sufficiently more normal than:

$$B(E) = \{s \in E : \text{ no } u \in E \text{ has } u \gg s\}.$$

Knowledge goes beyond evidence by the same mechanism, but its extent depends on where one actually is. I work mainly with K-normality (Goodman and Salow 2023, 104):

$$K_E(x) = B(E) \cup \{v \in E : v \succeq x\}.$$

One knows what is true throughout $K_E(x)$. Evidence is transparent: situations with the same evidence are evidentially accessible from each other. As Goodman and Salow note, "given transparency, K-normality entails KK" (2023, 107). The alternatives they consider, I- and KM-normality, validate their principle Margins and allow failures of KK; I return to them in §6.1. Discovering p at $\langle x, E\rangle$ yields $\langle x, E \cap p\rangle$. A question is a partition Π of E into discoverable cells; $\Pi(s)$ is the cell containing s.

Knowledge-based expected utility. Fix a prior Pr with full support. A decision problem is a finite set A of acts, each a function from states to utilities. At $\langle x, E\rangle$, KDT chooses an act maximizing $\Pr(u_a \mid K_E(x))$, with ties broken by a fixed order. Goldschmidt (2024) describes this conditionalizing form of KDT alongside Hawthorne and Stanley's proposal, on which known propositions receive probability 1; for present purposes the two agree.

The value of a free question. Let $\alpha_\Pi(s)$ be the act KDT chooses at $\langle s, E \cap \Pi(s)\rangle$, after answering Π in state s. The KDT value of answering Π before choosing is

$$V_x(\Pi, A) = \sum_s \Pr(s \mid K_E(x))\, u(\alpha_\Pi(s), s) \;-\; \max_{a \in A} \Pr(u_a \mid K_E(x)).$$

This is Good's comparison of acting now with the plan of looking and then acting, with $\Pr(\cdot \mid K_E(x))$ in place of the unconditioned probability. Call Π safe at x if $V_x(\Pi, A) \geq 0$ for every decision problem A.

Two conditions will matter.

Relative defeat at x for Π: some $s \in K_E(x)$ has $K_{E\cap\Pi(s)}(s) \not\subseteq K_E(x)$. In words, for all one knows, answering the question would leave open something one now rules out.

Gapped: every member of $K_E(x)$ is sufficiently more normal than every member of $E \setminus K_E(x)$.

3. Bjorn's second weighing

Goodman and Salow's (2023, §6) model of Bjorn is statist and comparable, and it violates collapse. Its states are triples $\langle w, y_1, y_2\rangle$: Bjorn's weight and the readings of two scales. With $\delta = (w - y_1)^2 + (w - y_2)^2$, one state is at least as normal as another iff its δ is no larger, and sufficiently more normal iff its δ is smaller by more than c^2. Goodman and Salow derive these relations from independent Gaussian errors (2023, 129). I take the standard deviation as the unit and, following BRN (§5, n. 18), set $c \approx 1.96$.

After the first reading r, Bjorn believes his weight lies in $[r-c, r+c]$. Under K-normality, in any situation with $\delta \leq c^2$ this belief is knowledge, and his ken is $\{\delta \leq c^2\}$. After a second reading r' he believes his weight lies within $c/\sqrt{2}$ of the average of the two readings. So he leaves open weights up to $c/\sqrt{2} + |r - r'|/2$ from r. As Goodman and Salow show for the case where the first reading is accurate, he "loses knowledge" whenever $|r - r'| > (2 - \sqrt{2})c$ (2023, 127). Since the readings may differ by up to $\sqrt{2}c$ for all he believes, he believes that the second weighing might defeat his knowledge. This is a failure of Preservation, the case BRN use against it. It is not rare: with Gaussian errors, the second reading falls in the Preservation-violating range with probability about 0.37, as BRN also estimate (§5, n. 18). Salow describes such failures as "relatively rare" (2025, 36); in the leading model of measurement they occur on more than a third of second weighings.

Ten-to-one bet. After the first weighing Bjorn is offered a bet: win 1 if his weight is within c of r, lose 10 otherwise. He knows the bet wins, so KDT takes it. A free second weighing is then offered. After the second weighing he would decline the bet whenever the readings differ by more than about $1.12c$, because his new ken gives the losing weights enough probability. Conditional on his present knowledge, this has probability about 0.05. In each such state he forgoes a payoff he now knows he would get. So KDT values the free weighing at about -0.05. At twenty-to-one the value is -0.12, and at a hundred-to-one -0.27. (These are numerical integrations over the model; see the appendix.) A fallibilist who uses the unconditioned probabilities also takes the ten-to-one bet: on her evidence it wins with probability 0.95, so its expected value is $0.95 - 10(0.05) = 0.45$. By Good's theorem the second weighing cannot have negative value for her.

The phenomenon is not confined to normal situations. At any situation after the first weighing there is a state compatible with Bjorn's knowledge in which the second reading would reopen a weight he now rules out (appendix, A.5). The finite-state Lemma 1 cannot by itself establish that claim in a continuous space, where a single state has probability zero. Appendix A.5 supplies the positive-measure extension. Thus some finite-stakes bet makes the second weighing harmful at each situation; the required stakes can vary with the situation. The moderate ten-to-one example is established for the normal situations specified above.

4. Where Good's theorem holds

The case instantiates a general pattern. Write M for any assignment of a set $M_F(s) \subseteq F$ to situations, used in place of K in the definitions of §2.

Lemma 1 (defeat is costly). If some $s_0 \in M_E(x)$ has $M_{E\cap\Pi(s_0)}(s_0) \not\subseteq M_E(x)$, then some two-act decision problem gives the question Π negative value.

Proof. Take a bet β paying 1 on $M_E(x)$ and $-L$ elsewhere, and declining, which pays 0. Now β has conditional expectation 1, so it is chosen. At s_0 the post-answer state contains a state outside $M_E(x)$, which has positive probability, so for large L the agent declines. She then gets 0 where β paid 1. Nowhere in $M_E(x)$ does she get more than 1. So $V \leq -\Pr(s_0 \mid M_E(x)) < 0$. □

Lemma 2 (laminar sufficiency). Suppose the post-answer states $\{M'(s) : s \in M_E(x)\}$ all lie within $M_E(x)$. Suppose also that they are reflexive ($s \in M'(s)$), transitive ($t \in M'(s)$ implies $M'(t) \subseteq M'(s)$) and nested (any two are disjoint or one contains the other). Then $V \geq 0$ for every decision problem.

This is the sufficiency half of a result of Geanakoplos (1989), as reported by Das (2023). I give a short proof in the appendix.

Lemma 3. Under statism and comparability, for each F the K-normality kens $\{K_F(s) : s \in F\}$ are reflexive, transitive and nested.

Theorem 1. Assume statism, comparability and K-normality. (a) A question is safe at x if and only if there is no relative defeat at x for it. (b) Suppose that for each $v \in E$ the question $\{p_v, E \setminus p_v\}$ is available, where $p_v = \{u \in E : \text{not } u \gg v\}$. Then every available question is safe at x if and only if $K_E(x)$ is gapped.

Part (a) combines Lemmas 1–3. Part (b) adds two observations. If the ken is gapped, no answer can reopen anything. If it is not gapped, the question about p_v reopens v. That is the next lemma, and it does not depend on which theory of knowledge one adopts.

Lemma 4 (non-gaps are defeasible). Assume statism. Let $m \in M_E(x)$ and $v \in E \setminus M_E(x)$, with not $m \gg v$. Then $m, v \in p_v$ and $v \in B(E \cap p_v)$.

Proof. $m \in p_v$ by hypothesis and $v \in p_v$ by irreflexivity. Nothing in p_v is sufficiently more normal than v, so $v \in B(E \cap p_v)$. □

Say that M leaves open what rational belief leaves open if always $B(F) \subseteq M_F(s)$. Knowledge on each of Goodman and Salow's three completions does, and so do justified true belief and the closure of knowledge under iteration.

Theorem 2 (only gapped states can be conditioned on). Assume statism. Suppose M leaves open what rational belief leaves open and the questions of Theorem 1(b) are available. If expected utility conditioned on M makes every available question safe at x, then $M_E(x)$ is gapped. A gapped set containing x contains $C(x)$: the states reachable from x by steps, each to a state that is not sufficiently less normal than the last.

The boundary in Theorem 1 is not the boundary of Preservation. At abnormal situations, knowledge can be defeated by evidence that contradicts one's beliefs, so KDT can disvalue a question even in models where Preservation holds without exception. Take three states a, b, c, in decreasing order of normality, with $a \gg b$ and $a \gg c$ but not $b \gg c$. In this

structure Preservation holds for every body of evidence and every discovery. At b one knows $\{a, b\}$. Learning that a is false, which is true at b, reopens c. Salow anticipates this pattern: a 'for all you know' version of Weak Knowledge Dominance "can support dogmatism even given Preservation" (2025, 33). Theorem 1 locates its expected-utility counterpart exactly.

Call E graded if $C(x) = E$ for every $x \in E$. Bjorn's evidence after one reading is graded, since a chain of small increases in δ connects any two states. So is the evidence in Flipping for Heads, where flips are counted until the first heads (BRN §5).

Corollary. (i) Under statism, collapse and comparability, every K-normality ken is gapped, so KDT makes every question safe. (ii) If E is graded and the questions of Theorem 1(b) are available, expected utility conditioned on any state that leaves open what rational belief leaves open makes every question safe at x only if that state is all of E.

Part (i) is the decision-theoretic counterpart of BRN's result that statism, collapse and comparability validate Preservation. Part (ii) says that in graded models the only belief-containing state one can condition on without making some free question harmful is one's evidence. Conditioning on evidence is what Salow calls the fallibilist response, on which "decisions should be governed by underlying probabilities rather than our knowledge" (2025, 23). Bjorn shows that the questions needed for (ii) can be as natural as a second reading.

I also checked the theorems by exhaustive enumeration. The enumeration covered all statist comparable structures with up to four states, every body of evidence, every situation and every partition: 64,004 situation–question pairs. Across all available partitions, a situation admitted some relative defeat exactly when its ken was not gapped. This is an existential claim over questions: a particular question can be safe at a nongapped ken. In each of the 3,306 cases of relative defeat, the decision problem built in Theorem 3 below gave negative KDT value. About 240,000 random decision problems without relative defeat gave no negative value.

5. The conditional repair

5.1 The rule

The natural expected-utility analogue of Conditional Knowledge Dominance compares two options by their expected difference on the hypothesis that they differ. Let $D_{a,b} = \{s \in E : u_a(s) \neq u_b(s)\}$. Let $K_E^D(x)$ be what one knows on the hypothesis D. Its K-normality form restricts evidential accessibility to D, as BRN's definition of conditional belief does (App. D.3):

$$K_E^D(x) = \{v \in E \cap D : \text{no } u \in E \cap D \text{ has } u \gg v\} \cup \{v \in E \cap D : v \succeq x\}.$$

Under statism, when $x \in D$, this is the ken one would have on discovering D; in every case it contains the belief one would have on discovering D. So it respects Salow's constraint that one is "entitled to believe q on the hypothesis that p only if discovering p would not require you to stop believing q" (2025, 33).

CEU. Prefer b to a if $\Pr(u_b - u_a \mid K_E^{D_{a,b}}(x)) > 0$. Looking before acting is one option; its payoff in state s is $u(\alpha_\Pi(s), s)$.

For Bjorn's bet CEU works as Salow's diagnosis predicts. Looking and betting now differ exactly where the second reading would make Bjorn decline. On the hypothesis that this happens, his conditional ken includes losing weights. With the actual readings perfectly accurate, at ten-to-one the bet's expected value on that hypothesis is -0.63. So CEU prefers to look. The dogmatic argument fails for the reason Salow gives: on the hypothesis that the evidence makes a difference, you do not know that your verdict is right.

5.2 Why the repair fails

For the displayed bet, the conditional repair uses states where looking and acting now pay the same. These are excluded from D, so conditioning on a difference carries information. That feature can be removed by a carefully chosen perturbation: if the present act and later policy differ in every state, $D = E$, conditional knowledge is ordinary knowledge, and CEU coincides with KDT. The following construction establishes such a case whenever there is relative defeat. No claim about the prevalence of ties in typical decision problems is needed.

Theorem 3. Assume statism, comparability and K-normality, and suppose conditional knowledge on the trivial hypothesis E is ordinary knowledge. If there is relative defeat at x for Π, then some three-act decision problem makes CEU strictly prefer acting now to answering Π. In it, KDT also gives Π negative value.

Construction. Let s_0 witness relative defeat. The acts are the bet β of Lemma 1, declining, and a perturbed bet $\beta_1 = \beta + \varepsilon\zeta$, where ζ is a large positive number N at s_0 and -1 elsewhere. Choose N so that ζ has positive expectation on $K_E(x)$; then β_1 is chosen now. After the answer, any post-answer ken that reopens something declines, given large L. Any post-answer ken that reopens nothing excludes s_0 (by transitivity, Lemma 3); there $\zeta \equiv -1$, so β beats β_1. So looking never leads to β_1, and its payoff differs from β_1's in every state. Hence $D = E$, and CEU compares the options on $K_E(x)$. At s_0, β_1 pays $1 + \varepsilon N$ while looking pays 0. Elsewhere looking gains at most ε. For small ε, β_1 wins. (Details in the appendix.)

For Bjorn at ten-to-one, take ζ positive on the states where the second reading would make him decline. CEU then prefers betting now by 0.05, the same margin by which KDT disvalues the weighing.

One might reply that the perturbation is trivial. Surely the rule should condition on the hypothesis that looking makes a significant difference. That reply does not help. For any fixed threshold θ, rescale the payoffs so that ε exceeds θ and raise L; the construction goes through, since nothing in it requires ε to be small once $N \Pr(s_0 \mid K_E(x)) > 2$ (A.6). More basically, the anti-dogmatic thought behind the repair is that "if the evidence you would discover tells strongly against your view, there is a good chance you're wrong" (Salow 2025, 36). In the perturbed problem, looking makes a difference in every state, including those where, by the agent's knowledge, the evidence misleads. Conditioning on

"it makes a difference" then removes nothing, and the knowledge-conditioned weighing that produced the dogmatic verdict returns unchanged.

So CEU is safe at most where KDT is safe. It is sometimes safe at fewer places. In the exhaustive search, CEU strictly preferred acting now in some problems at situations without relative defeat, where KDT is safe. In those problems the hypothesis D is false at x, and the conditional ken reopens possibilities that one's knowledge excludes.

5.3 Which conditional knowledge?

Even in problems with exact ties, CEU needs a choice that the normality framework does not settle. Without statism, BRN distinguish belief conditional on q (their definition restricts accessibility to q while keeping the current evidence's normality relations) from belief after discovering q. They show that the first can hold without the second (App. D.3). Their example is a variant of Flipping for Heads from Goodman and Salow (2023, 138). The coin is fair; q says it was flipped either once or more than n times. Normality comes from probability by BRN's threshold rule. I take $n = 4$ and $t = 0.9$, so that one initially believes, and in the actual state s_1 (heads first) knows, that there were at most four flips. On the hypothesis q one believes there was one flip. After discovering q one leaves five flips open.

Offer twenty-to-one on "at most four flips" and the free question whether q. KDT values the question at -0.53. Looking changes the payoff exactly on q, so $D = q$. On BRN's reading of conditional knowledge, the ken on D at s_1 is $\{s_1\}$, and CEU prefers betting now. On the discovery reading it is $\{s_1, s_5\}$, and CEU prefers looking. The same split arises for Salow's qualitative principle restricted to accuracy about the bet's proposition. With BRN's conditional belief, you believe that keeping your belief is better on the hypothesis that the question makes a difference, so Conditional Knowledge Dominance says not to look. In this example the repair works on the post-discovery reading and fails on the specified alternative reading. That is Salow's constraint, and outside statist models BRN's account of conditional belief rejects it. Their reason is not idle: conditional belief must make sense for hypotheses one could never learn, such as "p, but I will never learn that p" (App. D.3). A defender of the repair can keep both only by letting the practical role of conditional knowledge diverge from its doxastic role exactly in the non-statist cases that BRN take to be typical.

6. Objections and replies

6.1 "This is self-trust, not evidence"

The objection says that KDT disvalues questions because the agent distrusts her future self, not because she distrusts the evidence. On the Margins completions (I- and KM-normality), statism, comparability and Margins "together entail no defeat" (Goodman and Salow 2023, 128). There, by Lemma 4, the reopened possibility lies in the agent's own present ken at m, so the relative defeat in Theorem 2 is a failure of KK at x. Such failures harm the empty question too. By Lemma 1 with the trivial partition, a KDT agent without KK at x pays to decide now rather than later on the same evidence. This kin of Williamson's unmarked clock is discussed by Ahmed and Salow (2019) and Das (2023).

For the Margins views I concede the objection: what fails there is deference to one's later self, not openness to evidence.

For K-normality the objection fails. KK holds, so by Lemmas 2 and 3 the empty question always has nonnegative value. Zero is not guaranteed: at an abnormal situation the present ken may contain states with strictly smaller kens, whose state-dependent choices can outperform one fixed act. For example, take a more normal than b, with $a \gg b$, a uniform prior, and acts paying $(1, 0)$ and $(0, 2)$ at (a, b). At b the present best value is 1; with no new evidence, the state-dependent choice plan pays $(1, 2)$ and has value 1.5. The empty question therefore has value 0.5. The relevant contrast is absence of a negative value, not universal equality to zero. The negative value of Bjorn's second weighing therefore comes from the evidence. The reading he would see destroys knowledge he now has, in a situation where his knowledge is fully introspective. The structure is the classic dogmatist's. I know the bet wins. In some states compatible with what I know, the evidence will make me unable to rely on that. So I should avoid the evidence. Under Theorem 1's assumptions, some available question has this structure exactly where knowledge is not gapped. So the dilemma for graded inductive knowledge is this. On the KK-friendly completion, KDT avoids evidence. On the KK-unfriendly completions, it avoids deferring to itself.

6.2 "Trivial once stated: knowledge is treated as certainty"

The pieces are indeed familiar. Lemma 1 is an ordinary bet, Lemma 2 is Geanakoplos's sufficiency condition, and Corollary (i) parallels BRN's result. Three things are new. The first is the exact location of the boundary, which includes abnormal situations whose kens extend below the belief set. The second is Theorem 2's completion-independence: no belief-containing state other than a gapped one can be conditioned on, so no refinement of knowledge (iterated knowledge, justified true belief) escapes. The third is Theorem 3. Harman's and Kripke's discussions explain why knowledge-as-certainty invites the puzzle. They do not show that the leading conditional remedy fails in expected-utility form. Nor do they show that, on graded normality, conditioning on evidence is the only way out.

6.3 "KDT is meant for low stakes"

Stakes-sensitive variants condition on higher-order knowledge when stakes rise (Goldschmidt 2024 surveys them). Under K-normality with transparency, knowledge iterates, so such variants coincide with KDT here. The stakes needed are moderate in any case. Bjorn's case needs only ten-to-one, where the fallibilist bets too.

6.4 "Statism is the exception"

BRN argue that statism fails in general. Three replies. Bjorn's model, which carries §3, is statist. Lemma 4 survives without statism in BRN's simple probability structures. If m_0 is the most probable member of p_v, then p_v consists of the states no more probable than m_0. "Not $m_0 \gg v$" says that the probability, given $E \cap p_v$, of the states more probable than v is below threshold. By BRN's characterization of belief (Prop. 5) this puts v in

$B(E \cap p_v)$. And non-statist models only add failures, as §5.3 shows: discovering what one already knows can destroy knowledge (Goodman and Salow 2023, §8).

6.5 Encroachment

Salow (2025, 24 n. 3) reports Hawthorne's (2004, 181) suggestion that, given Good's theorem, the puzzle is handled by pragmatic encroachment of the kind on which encroachment occurs whenever the difference between maximal and actual probability affects one's preferences over the available options. Salow objects that encroachment would not be confined to failures of Preservation (36). Theorem 2 makes the cost exact. To guarantee safety for every decision problem and every question in the theorem's class on graded evidence, a factive belief-containing state must shrink to the evidence. This does not show that the mere availability of a second scale destroys all inductive knowledge at every fixed stake level. The universal guarantee quantifies over decision problems as well as questions.

6.6 "Good's theorem is not sacrosanct"

Ahmed and Salow (2019) argue that Good's theorem may fail. What is rational, on their view, is to make one's act depend on the answer to a question, and looking first is rational exactly when it is a way of doing that. Their counterexamples involve inexact, non-partitional observation. Bjorn's second weighing is not of that kind. The reading is observed exactly, the evidence is transparent, and the possible readings partition the states. Looking is therefore a way of making his act depend on the answer to a question, which is the advice they endorse. So the case does not belong among their counterexamples. The challenge is therefore to explain why avoidance in this partitional measurement case is rational, rather than assimilating it without further argument to their non-partitional examples.

6.7 Carter and Hawthorne's inquiry-based dogmatism

Carter and Hawthorne (2024) consider Ex Ante Dogmatism, the recommendation that one avoid gathering evidence bearing on what one knows. They ask how a view on which inquiry aims at knowledge can resist it. The present result is about expected utility, not the aims of inquiry. But it locates exactly where a knowledge-conditioned decision rule recommends ex ante avoidance (non-gapped kens), and, for the finite theorem's question class, it shows that every proper factive belief-containing ken in graded evidence has some unsafe question.

7. Conclusion

In finite statist, comparable models with K-normality and the specified questions available, expected utility conditioned on knowledge makes every free question safe exactly where knowledge is gapped. Any factive state that leaves open what rational belief leaves open needs the same condition. On graded normality, which Goodman and Salow defend

for measurement, that condition means not going beyond the evidence. The expected-utility form of Salow's Conditional Knowledge Dominance does not escape: it disvalues free evidence wherever the unconditional rule does, and in non-statist models its success depends on identifying conditional with post-discovery knowledge. What has not been shown is that no other way of letting all-out attitudes guide graded choice could preserve Salow's dissolution. The result places a precise burden on this conditionalizing form of knowledge-first decision theory. It must weaken the universal safety requirement, restrict its model or question class, or change how knowledge controls numerical choice. Collapse is one sufficient restriction, and evidence-based probabilities are one alternative; the results do not establish an exhaustive trilemma for every knowledge-first theory.

Appendix: proofs and computations

A.1 Lemma 2. Let $\mathcal{F} = \{M'(s) : s \in M\}$, $M = M_E(x)$. By nestedness and reflexivity the inclusion-maximal members of $\mathcal{F}$ partition M. For a maximal $C = M'(c)$, let a_C be the act chosen on C. I show by induction on $|C|$ that $\Pr(u(\alpha(s), s) \mid C) \geq \Pr(u_{a_C} \mid C)$. By transitivity every $s \in C$ has $M'(s) \subseteq C$. The states with $M'(s) = C$ choose a_C. The others lie in the maximal proper sub-kens $D_1, \ldots, D_k$, which are disjoint members of $\mathcal{F}$ satisfying the hypotheses. By induction, on each D_j the plan does at least as well as the best act on D_j, and so at least as well as a_C. Summing over the parts of C and then over the maximal sets, the plan's expectation on M is at least $\sum_C \Pr(C \mid M) \max_a \Pr(u_a \mid C) \geq \max_a \Pr(u_a \mid M)$. □

A.2 Lemma 3. Reflexivity holds because $s \succeq s$. For transitivity, let $t \in K_F(s)$. If $t \in B(F)$, then every $v \succeq t$ is in $B(F)$: if $u \gg v$, chaining gives $u \gg t$. So $K_F(t) = B(F) \subseteq K_F(s)$. If $t \succeq s$, then $\{v : v \succeq t\} \subseteq \{v : v \succeq s\}$. For nestedness, by comparability the sets $\{v : v \succeq s\}$ are linearly ordered by inclusion, and each ken is their union with $B(F)$. □

A.3 Theorem 1. (a) Answers in different cells give disjoint kens. By Lemma 3 the post-answer kens are reflexive, transitive and nested. Without relative defeat they lie within $K_E(x)$, so Lemma 2 applies. Conversely, use Lemma 1. (b) Suppose the ken is gapped, $s \in K_E(x) \cap c$, and $v \in c \setminus K_E(x)$. Then $s \gg v$, so $v \notin B(E \cap c)$. Also not $v \succeq s$, since otherwise $v \succeq s \gg v \succeq v$ gives $v \gg v$. So $K_{E \cap c}(s) \subseteq K_E(x)$. If the ken is not gapped, Lemma 4 gives relative defeat, because $B(E \cap p_v) \subseteq K_{E \cap p_v}(m)$. □

A.4 Theorem 2. Lemmas 4 and 1 give the first claim. If M is gapped, $m \in M$ and not $m \gg v$, then $v \in M$; so M is closed under the steps that generate $C(x)$. □ Corollary (i). Let $m \in K_E(x)$ and $v \notin K_E(x)$. Then not $v \succeq x$, and some $u \gg v$. If $m \succeq x$, comparability and collapse give $x \gg v$, hence $m \gg v$. If $m \in B(E)$, then $v \succeq m$ would give $u \gg m$, so m is strictly more normal than v and collapse gives $m \gg v$. □

A.5 Bjorn and positive measure. After the first reading, use coordinates $\mu = (e_1 + e_2)/2$ and $d = e_1 - e_2$, so $\delta = 2\mu^2 + d^2/2$. For the numerical model assume the proper conditional Gaussian measure with independent $\mu \sim N(0, 1/2)$ and $d \sim N(0, 2)$, corresponding to the swamped-prior treatment of weight. The measurement likelihood alone would not fix this conditional distribution without that prior assumption. Conditional on a second reading, d is fixed and the post-answer ken has $|\mu| \leq \max(c/\sqrt{2}, |\mu_s|)$.

Let $R = \max(c^2, \delta(x))$, so the current ken is the disk $M = \{\delta \leq R\}$. Choose an interior point with $\mu_s = 0$ and $d(s)^2/2$ strictly between $R - c^2$ and R. Its post-answer ken reaches beyond M, since its maximum δ is $c^2 + d(s)^2/2 > R$. The inequalities are strict. Hence a sufficiently small open neighborhood U of that point stays inside M and has post-answer conditional probability of M^c bounded below by some $\eta > 0$. Gaussian density makes U have positive current probability. A bet paying 1 on M and $-L$ outside it is therefore declined throughout U when $L > (1-\eta)/\eta$. It is accepted now; subsequent choice never pays more than 1 inside M. Its value of looking is at most $-\Pr(U \mid M) < 0$. This is the continuous argument that the finite singleton proof cannot supply.

For the stated normal-situation figures, the bet is instead on $|e_1| \leq c$. Integrating the Gaussian density over $\delta \leq c^2$ gives the probabilities of declining after the second reading. These figures are numerical approximations, not premises of the finite theorems.

A.6 Theorem 3. Let $P = \Pr(\cdot \mid K_E(x))$. Pick N with $NP(s_0) > 2$. Pick L so large that every post-answer ken containing a state outside $K_E(x)$ declines both bets. Pick $\varepsilon > 0$ so small that $\beta_1 \neq 0$ everywhere. Then β_1 is chosen now, since its expectation is $1 + \varepsilon P(\zeta) > 1$. Looking never leads to β_1, so $D = E$ and CEU uses $K_E(x)$. Then $P(u_{\beta_1} - u_{\text{look}}) \geq P(s_0)(1 + \varepsilon N) - \varepsilon > 0$. □

Computations. Python scripts enumerate all statist structures satisfying Goodman and Salow's constraints: comparable ones with up to four states (up to five for collapse) and non-comparable ones with up to three. For each they compute kens, post-answer kens, KDT and CEU choices with random full-support priors and the constructed problems. Bjorn's values are computed by numerical integration. Flipping for Heads is computed exactly, truncated at 60 flips.

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Flattening and Counterfactual Independence

Abstract. Dorr and Mandelkern characterize the logic obtained by adding Flattening to Stalnaker's C2 and argue for its appeal as a principle of natural-language conditionals. I develop a counterfactual challenge using matched cases of independence and avoidance. In explicit models the cases agree on first-degree counterfactuals while differing on a nested conditional that Flattening would identify with an unnested one. On a scenario-relative reading, the nested sentence follows the dependencies of the scenario reached by its outer antecedent. That reading supplies the proposed counterexample. The formal separation is clear; the linguistic conclusion depends on judgments that have not been independently tested. Context-sensitive and sequence-based replies remain live, and I specify what they must explain. The result is a focused test of Flattening's natural-language motivation, not a new axiomatization or a general refutation of sequence semantics.

1. Introduction

Van Fraassen (1976) evaluated conditionals at sequences of worlds: "if p, then q" is true at a sequence iff q is true at the first tail of the sequence at which p is true. Dorr and Mandelkern (2026) have determined the logic of this semantics. With sequences of length ω it is C2.FS, Stalnaker's (1968) logic C2 plus two schemas, Flattening and Sequentiality. With sequences of arbitrary ordinal length it is C2.F, which adds Flattening alone:

$$\text{(Flattening)} \qquad p > (pq > r) \;\leftrightarrow\; pq > r.$$

They argue that Sequentiality is probably invalid and that Flattening has strong prima facie appeal. Their case has four strands.

Similarity. Similarity semantics cannot validate Flattening, but that tells against similarity, not against Flattening. Their own counterexample, built on a toy example of Lewis's, is hard to hear "except by doggedly holding in mind the Lewisian interpretation" (2026, 19–20).

Test pairs. Sentences that Flattening makes equivalent feel equivalent. At most the nested member feels redundant: "the first antecedent feels like it's doing nothing", which validity would explain (20). "We have checked many instances of Flattening, in both the indicative and subjunctive mood, and have not found clear counterexamples." (20). Felt inequivalences that come from re-indexing tense or anaphora do not count, since parallel pairs would refute Contraction too (20–21).

Import-Export. Import-Export, $p > (q > r) \leftrightarrow pq > r$, collapses C2 into the material conditional, as they report from earlier work (21), and it has subjunctive counterexamples. So a unified theory of "if" should not validate it, and should explain its indicative appeal by the interaction of "if" with mood. "But matters are different for Flattening, which appears valid for both indicatives and subjunctives." (22).

Cautious cousins. Cautious Monotonicity and Cautious Transitivity are theorems of C2, while their incautious versions yield the material conditional. By analogy, "we might

naturally regard Flattening as a plausible cautious cousin of the implausible, incautious Import-Export principle" (23).

They conclude that C2.F has "strong prima facie appeal as (at least part of) the logic of the natural language conditional" (24). They record two reservations. Flattening yields the modal schemas 4 and H for the necessity operator defined from the conditional, and both schemas are contested (23–24). And no one has an intuitive semantics that explains Flattening's validity: "we leave it as an open problem for proponents of C2.F" (24).

I argue that a scenario-relative reading of nested counterfactuals gives a serious challenge to Flattening. Two matched cases receive different nested judgments while their unnested partners receive the same judgment. The proposed reading evaluates the inner supposition using the dependencies of the scenario reached by the outer supposition. If that is a genuine reading of the bare sentences under a stable interpretation of the conditional, Flattening fails on that reading. Whether ordinary speakers accept this interpretation consistently is a further question; the examples here are judgments and models, not experimental results. I make no claim against Flattening for indicatives, leave 4 and H open, and use Dorr and Mandelkern's logical results throughout.

The method needs one preliminary. Section 2 notes that C2.F adds nothing to C2 in its unnested fragment, probabilities included. So any case for or against Flattening, theirs or mine, must rest on judgments about nested conditionals. The live questions are whether those judgments are bare or enriched with particles such as "still" or "even", and whether one interpretation is held fixed across them.

Dorr and Mandelkern give one counterfactual counterexample, which depends on a similarity metric, and they find it weak. The increment of this paper is as follows. (i) A counterexample of a different kind. It is a bare instance of the rule form of Flattening, in the right-nested fragment with conditional-free antecedents where their test pairs lie, and it is driven by counterfactual independence rather than by a metric. A matched control distinguishes several readings that could carry the verdict. (ii) A diagnosis of the explanatory burden on contextual replies, without claiming to eliminate every such reply. (iii) An explanation of why their subjunctive test pairs pass, and why their redundancy datum and cautious-cousin analogy do not tell between Flattening and its denial. (iv) The consequence that their own argument about Import-Export applies to Flattening. Two further observations are small: the conservativity point above, and the fact that the cases leave C2 plus both "crashing" halves of Flattening untouched. Several formal facts I use are reformulations of their results, and I credit them as such.

For readers outside conditional logic, the stake is this. Flattening looked like the safe part of Import-Export, the principle that supposing p and then a strengthening of p comes to the same as supposing the strengthening outright. If this reading is correct, a unified semantics must explain why these counterfactual suppositions combine in this way. The argument does not independently establish a contrast with indicative logic. A counterfactual supposition made inside a supposed scenario answers to that scenario's own dependencies. That matters to anyone who reasons with nested counterfactuals about causal structure.

Section 2 sets out the framework. Section 3 gives the minimal pair and its control, and Section 4 the deterministic case and a general theorem. Section 5 answers objections. Section 6 explains the test pairs, and Section 7 draws the consequences for the logic of "if".

2. Framework

I follow Dorr and Mandelkern's order semantics (2026, §2.1). An order frame assigns each world w a set $R(w)$ of accessible worlds containing w, together with a well-order $<_w$ of $R(w)$ in which w comes first. $p > q$ is true at w iff either $R(w)$ contains no p-world or q is true at the $<_w$-first p-world. C2 is sound and complete for these frames (their Theorems 2.4–2.5). It contains Conditional Excluded Middle (CEM), $p > q \vee p > \neg q$, and Normality. Possibility is defined by $\Diamond p := \neg(p > \neg p)$. I write $f(w, X)$ for the $<_w$-first world of $R(w)$ in X.

The two directions of Flattening are Cautious Importation, $p > (pq > r) \to pq > r$, and Cautious Exportation, $pq > r \to p > (pq > r)$. Their special cases with $r = \bot$ are Crashing Cautious Importation and Crashing Cautious Exportation. In C2 the latter is equivalent to 4, and the former yields H (§4, 18–19). Flattening also has a rule form: if $\vdash q \to p$, then $\vdash p > (q > r) \leftrightarrow q > r$ (18). Flattening is valid on a frame iff the frame is flat: transitive, and such that whenever x is accessible from w, $<_x$ orders the worlds after x in $<_w$ as $<_w$ does (Theorem 4.3). In selection terms this is

$$\text{(Flat)} \qquad f(f(w, Y), X) = f(w, X) \quad \text{whenever } X \subseteq Y,$$

which is their condition restated, and which they state themselves for metric spaces (24). C2.F is also complete for linear frames, where each world's order is a "single master order" with that world's predecessors deleted (Theorem 4.5; 27). In sequence semantics a tail's order is the rest of the sequence.

Observation 1. Call a conditional first-degree if its antecedent and consequent contain no conditionals. In a finite order model, any Boolean combination of first-degree conditionals true at a world is true at a world of some finite linear order model. Any probability assignment to such sentences realized by a finite probabilistic order model is also realized by a finite flat one.

Proof. A first-degree sentence's truth value at w depends only on $<_w$ and the valuation. Form a master order by listing the worlds outside $R(w)$ and then $R(w)$ in the order $<_w$, and let each world's order be the master order from that world on. The resulting frame is linear, and w keeps its order. For probabilities, take one such linearization for each world w, form their disjoint union (which is flat), and give the copy of w at the head of its own linearization the probability w had. □

So data about unnested counterfactuals, whether categorical or probabilistic, cannot tell against Flattening without telling against C2. They cannot tell in its favor either. (I checked the construction exhaustively on three-world frames and on random four-world frames.)

3. A bare minimal pair

> The Reunion. Ann and Bob studied together and have not been in touch for years. Both missed last Saturday's reunion. Ann has worked in Australia since January and never considered flying back. Bob lives near the venue and meant to come, but his babysitter cancelled that afternoon. Nothing that would have brought either of them would have affected the other.
>
> Avoidance. As in the Reunion, except that Bob has avoided Ann for years. The organizers posted the list of guests in advance, and Bob checked it. Had her name been on it, he would have stayed home.

Consider:

(1) If at least one of them had come, then if Ann had come, Bob would have been there.

(2) If Ann had come, Bob would have been there.

With A for "Ann came" and B for "Bob came", (1) is $(A \vee B) > (A > B)$ and (2) is $A > B$. Since $\vdash A \rightarrow (A \vee B)$, the rule form of Flattening makes them equivalent. Each direction of the biconditional is an instance of one half of Flattening. The sentences contain no "still", "even" or "too". The outer antecedent says "at least one", which has the truth conditions of the inclusive disjunction "Ann or Bob or both", so the formal instance has exactly that antecedent.

The judgments are these. (2) is false in both cases: had Ann come, Bob's babysitter would still have cancelled. In both cases, had at least one of them come, it would have been Bob, since his coming needed only a babysitter who kept her word while Ann's needed an intercontinental trip she never considered. In the Reunion, (1) is true: had Bob come, Ann's turning up as well would have changed nothing for him. In Avoidance, (1) is false: had Bob come, Ann's name on the list would have kept him at home. The argument is then:

(P1) (2) is false in the Reunion and false in Avoidance.

(P2) (1) is true in the Reunion and false in Avoidance.

(P3) Under any single interpretation of the sentences on which Flattening is valid, (1) and (2) have the same truth value in each case.

(C) So either Flattening fails under the interpretation on which (P1) and (P2) are correct, or in the Reunion (1) and (2) are not interpreted by a single interpretation.

(P3) is the rule form. The weight falls on (P2) and on the second disjunct of (C), which I take up in Section 5.1.

The control does a specific job. It does not show that the Reunion verdict is right. It identifies which reading of (1) carries the verdict. The two cases agree on every unnested counterfactual about who came. In both, the possibilities rank in the same order from actuality: neither came, then Bob alone, then Ann alone, then both. Now consider the readings a defender of Flattening might offer. On the exclusive reading, "if Ann had come" means "if it had been Ann (rather than Bob)", that is, $A \wedge \neg B$. On the additive reading it means "if Ann had come as well", that is, $A \wedge B$. Each of these entails $A \vee B$. So under Flattening, (1) on either reading reduces to an unnested counterfactual, and that

counterfactual has the same value in both cases. This holds for every construal of the inner antecedent that entails the outer one. The same is true without Flattening: on the exclusive reading (1) is false in both cases, and on the additive reading it is true in both. On the universal reading of the outer antecedent ("whichever of them had come"), (1) is false in both, because if Ann had been the one to come, Bob would not have been there.

Among these readings, the scenario-relative reading yields the contrast. On it, the outer antecedent selects the scenario in which Bob came, and "if Ann had come" is then evaluated from that scenario according to what depends on what there. Flattening fails on that reading. I computed every one of these predictions in explicit four-world models. Dorr and Mandelkern's tense and anaphora pairs involve re-indexing of this kind: the inner occurrence of an expression is construed differently from the outer one. The absence of an overt change in tense or reference makes their specific re-indexing diagnosis less immediate here. It does not rule out pragmatic shifts driven by the different background stories.

Is (P2) right? The Avoidance verdict is the one Flattening predicts, so the question is the Reunion verdict. Given CEM, a proponent of C2.F must hold that in the Reunion this is true: "If at least one of them had come, then if Ann had come, Bob would not have been there." That says that in the scenario where the babysitter kept her word and Bob came, Ann's arrival from Australia would have kept him away. Nothing in the case supports that, and in Avoidance the corresponding sentence is true precisely because something does. The judgments are mine, and they are testable. A two-by-two design that crosses case (Reunion or Avoidance) with sentence ((1) or (2)) separates the hypotheses. Flattening predicts no interaction. I predict that (1) changes value across the cases while (2) does not. A Contraction row, "If Ann had come, then if Ann had come, Bob would have been there", should pattern with (2) in both cases, as C2 requires. That row checks that nesting by itself does not produce the effect.

4. Locality and a general form

The Reunion relies on judgments about a babysitter and a trip. The next case puts the inner judgment on a different footing.

> The Buttons. At noon, Ann in Sydney and Bob in London could each have pressed a button. Neither did. Ann's finger was already moving and she pulled back at the last instant; Bob had firmly decided not to. The world is deterministic, and nothing that happened in either city in the final instant before noon could affect the other city at noon, since no signal could cover the distance in time.

(3) If at least one of them had pressed, then if Bob had pressed, Ann would have pressed.

(4) If Bob had pressed, Ann would have pressed.

The pair is an instance of the rule form, since $\vdash B \rightarrow (A \vee B)$. Had at least one of them pressed, it would have been Ann, whose press needed only a smaller departure from actuality. (4) is false: a departure confined to Bob's last instant leaves Sydney at noon as it actually was. For the same reason (3) is true. From the scenario in which Ann pressed, a departure confined to Bob's last instant leaves Ann's press in place. Both verdicts follow

if we adopt one counterfactual selection principle, applied once at actuality and once at the scenario the outer antecedent reaches:

(Locality) At a world with deterministic laws under which no influence travels faster than light, the supposition that a local event had occurred, realized by a departure confined to that event's immediate past, leaves unchanged every region that no influence from that past could reach.

Locality is motivated by local-intervention readings and by suitable local-miracle versions of Lewis's (1979) approach. Relativistic no-signalling alone does not entail it: the antecedent must be realized by the stipulated local departure, rather than by backtracking changes to common causes. A semantics that rejects that selection policy can reject the argument from Buttons. The Buttons involve no chance, so they are not touched by doubts about "would" counterfactuals with chancy consequents.

Suppose instead that the case is symmetric, so that it is unsettled which of them would have pressed. Then the following result applies.

Theorem 1. Let X be any sentence with $\vdash A \to X$ and $\vdash B \to X$. In C2 plus Cautious Importation, and also in C2 plus Cautious Exportation,

$$\neg(A > B),\ \neg(B > A),\ X > (A \vee B) \ \vdash\ X > \big[(A \wedge (B > \neg A)) \vee (B \wedge (A > \neg B))\big]. \qquad \text{(D)}$$

Proof. With Cautious Exportation: CEM turns $\neg(A > B)$ into $A > \neg B$, and the rule form gives $X > (A > \neg B)$. Likewise $X > (B > \neg A)$, and Normality combines these with $X > (A \vee B)$. With Cautious Importation: the rule form gives $X > (A > B) \to A > B$, so $X > \neg(A > B)$ follows by CEM, and then $X > (A > \neg B)$ by Normality. The rest is as before. □

The proof follows Dorr and Mandelkern's derivation of H (18–19), and it uses CEM essentially. In C2, given $\Diamond A$, the premise $\neg(A > B)$ is equivalent to $A > \neg B$, so the premises are ordinary would-nots. The theorem shows that whichever way the tie is resolved, Flattening carries the actual world's would-nots into the supposed scenario. On either resolution, one of two bare instances of the shape of (3)–(4) fails. Using this against Flattening still requires judging the nested conclusion (D) false in the symmetric Buttons. That judgment is again Locality, applied at the scenario reached. (D) is true in Avoidance, as it should be. I checked Theorem 1 with the Z3 solver on every order frame with at most five worlds that satisfies Dorr and Mandelkern's frame condition for either half (Lemma 4.2 and note 24).

5. Objections and replies

5.1 A shift of context

The defender's first move is to say that Flattening, like Contraction, concerns a single interpretation, and that the outer antecedent of (1) shifts the interpretation of the inner conditional. Then (1) and (2) would not form an instance.

Context-sensitive replies must be taken seriously. Von Fintel (2001) develops a dynamic account of counterfactual discourse, and Roberts (1989) studies modal subordination. A

difference in background dependencies can affect interpretation even when the sentences have the same words. There is therefore no valid inference from identical wording to the absence of a context shift.

One possible reply makes the shift persist. It predicts that after the nested Reunion sentence is accepted, the unnested sentence should be understood with the same adjusted background. A useful diagnostic is the continuation: "If at least one of them had come, then if Ann had come, Bob would have been there. But if Ann had come, he would not have been there: his babysitter had cancelled." On my judgment this discourse is coherent, suggesting that the outer supposition does not permanently change the background for later assertions. That is evidence against this particular persistent-shift explanation, conditional on the judgment. It does not rule out temporary or scope-sensitive accommodation.

A temporary shift could preserve the conditional operator's Flattening law at a fixed index while explaining why the two natural-language occurrences use different contextual parameters. Such an account would block the proposed inference from the sentence pair to a formal counterinstance. It would need to explain why the temporary background preserves Bob's presence in Reunion but not in Avoidance, and why the proposed Contraction control behaves differently. The scenario-relative account supplies one explanation directly through the dependencies of the selected world. Calling an alternative contextual does not make it mistaken, but it must offer comparably precise predictions.

This leaves a genuine choice between analyses. If the contrast is part of the compositional interpretation of a fixed counterfactual operator, it challenges Flattening. If it comes from an independently motivated parameter shift, it challenges a simple argument from apparent sentence equivalence to the operator's validity. Neither reading can be settled merely by redescribing the shift as semantic or pragmatic. A reader study with context controls would strengthen the linguistic case; none has been carried out here.

5.2 The sequence theorist

In sequence semantics, conditional-free sentences are settled by a sequence's first member. So two indices can agree on every categorical fact and differ on conditionals. Dorr and Mandelkern describe, without endorsing it, a gloss on which conditionals typically answer "non-factual" questions, and they credit a recent picture along these lines to Khoo (2022) (2026, 13). Sequence semantics has also been applied in recent work that they list, including work on counterfactual probability (Schultheis 2023; 2026, 3 n. 4). On this view, the index that "had at least one of them come" reaches is a tail. It shares the Bob scenario's categorical facts but carries the remainder of the actual ranking, so (1) is false there.

I grant that sequence models can represent the Reunion this way, and I do not claim to refute the view. My aim is to make exact what it must deny. The metaphysics of factual and non-factual questions is idle here, because the rule form settles the matter: whatever conditionals answer to, the sequence theorist must assign (1) the value of (2) in both cases. So they must deny the contrast in (P2), or relocate it into a shift, which was the topic of Section 5.1.

The Buttons sharpen the cost. The sequence theorist accepts Locality at actuality, since that is what makes (4) false, and nothing in the principle singles out the actual world. Applied to the Ann-pressed scenario taken as actual, Locality makes "if Bob had pressed, Ann would have pressed" true. Reached by supposition, the same scenario must make it false. So the theory must hold that whether Ann's press would have survived Bob's depends not on the scenario's laws and local facts but on the supposition through which the scenario was entered. The nested counterfactual it must then accept reads: had at least one pressed, then had Bob pressed, Ann would not have pressed. That asserts a counterfactual connection between spacelike-separated events in a world whose laws forbid influence between them.

The same cost appears in causal modelling. In a structural-equation model (Galles and Pearl 1998; Halpern 2000), a nested counterfactual whose antecedents set variables is evaluated by intervening and then intervening again in the resulting model. Such models do not by themselves fix how a disjunctive antecedent is realized. Once it is realized, as here by setting the babysitter variable, the result is a fully specified model of the Bob scenario. In that model, intervening on Ann's variable leaves Bob's value as it is, and the verdict does not depend on how the model was reached. Flatness requires the opposite verdict.

5.3 Disjunctive and alternative-sensitive antecedents

Some hear "had Ann or Bob come" as "had Ann come, and had Bob come". Together with the substitution of logically equivalent antecedents, that simplification yields antecedent strengthening (Ellis, Jackson and Pargetter 1977), so a proponent of C2.F cannot rest on it. Experimental work reports support for selection-based readings (Ramotowska et al. 2025). In any case, Section 3 showed that the universal reading predicts no contrast, so it cannot be the reading behind (P2). Truthmaker and alternative-sensitive semantics refuse to identify $(A \vee B) \wedge A$ with A (Fine 2012; Santorio 2018; see Ciardelli, Zhang and Champollion 2018 on a two-switch case). Indefinites such as "at least one of them" raise alternatives just as disjunctions do. But the rule form, which is the form of Dorr and Mandelkern's own tests, needs only that the inner antecedent entail the outer. The control is neutral between ways of individuating antecedents, because whatever alternatives "at least one of them" raises, it raises them in both cases.

6. Why the test pairs pass

Let v be the scenario selected for the outer antecedent p. The two sides of an instance diverge iff the first pq-scenario reached from v differs, on what r concerns, from the first pq-scenario reached from actuality. Suppose each scenario ranks alternatives by its own dependencies. Then robust divergence needs three things. (a) The outer antecedent can be realized in several ways. (b) The selected realization settles a matter M that the inner antecedent leaves alone, so that supposing pq from inside v keeps M. (c) The consequent concerns M, and supposing pq from actuality settles M differently. When the ordering is a metric, divergence can arise without (b), from geometry alone. That is the case of their line example. It is hard to hear because nothing in the case makes 13 inches "nearer" to 11 than 8 is except the metric, and they were right to be unmoved by it.

Their subjunctive pairs fail (b) or (c). In "If he had gotten an espresso, then he would have had a fit if he'd gotten an overextracted one." (20), the inner antecedent fixes what the consequent depends on, however the outer one is realized. The same holds for the south of France and the soaked match. (Their conflagration pair is indicative, and I return to indicatives below.) Interventionist semantics shows why spot checks rarely find counterexamples. Intervening on a set of variables and then on a superset gives the same model as intervening on the superset directly, so instances whose antecedents are conjunctions of variable settings fail (a). The pair (1)–(2) meets all three conditions.

This also answers the redundancy datum. Where (b) or (c) fails, the outer antecedent of the nested member makes no difference in the context, so it "feels like it's doing nothing". That is predicted by contextual equivalence without logical equivalence. In (1) the outer antecedent does work: it is what makes the sentence true.

The cautious-cousin analogy fails for a structural reason. Cautious Monotonicity and Cautious Transitivity involve a single supposition. Every conditional in them is evaluated by the ordering at one world, which is why they hold on every order frame. Flattening involves iterated supposition. It relates the ordering at a world to the ordering at the world selected from it, which is why it needs a frame condition. The safety of the cautious principles comes from their staying within one ranking, so it does not transfer. (Nonmonotonic logic calls the conjunction of Cautious Monotonicity and Cut "cumulativity", and C2 already has that property. The property at issue here is different.)

Call a supposition operation iteration-invariant if supposing Y and then some $X \subseteq Y$ gives the same result as supposing X outright. Imaging a probability function on X moves each world's probability to $f(w, X)$ (Lewis 1976), and it is the standard model of counterfactual supposition, as conditioning is of indicative supposition (Eva, Shear and Fitelson 2022 compare these and other approaches). A reformulation of (Flat): on a finite order frame, Flattening is valid iff imaging is iteration-invariant for every probability function. (Point masses are imaged to $f(f(w, Y), X)$ and $f(w, X)$ respectively.) In the theory of belief change, the same condition on iterated revision is the first Darwiche–Pearl postulate: if $\alpha \subseteq \beta$, then $(\kappa * \beta) * \alpha = \kappa * \alpha$ (Darwiche and Pearl 1997, as stated by Hunter and Delgrande 2011, 297). That literature contrasts revision, which takes in information about a static world, with update, which projects each state to the state that would result from a change (Hunter and Delgrande 2011, 273). Imaging moves each world separately, much as update does, and the cases show why iteration invariance is unwelcome there. A world's ranking of what lies beyond a scenario need not be that scenario's own ranking. In flat frames, any two worlds that both reach a world v must order alike every pair of worlds that both place after v (an immediate consequence of flatness). Dorr and Mandelkern's master-order picture makes this vivid. In the Reunion, actuality ranks Ann-alone ahead of both-came, because Bob's staying home is held fixed. The Bob scenario ranks them the other way.

7. Which logic, indicatives, and the unity of "if"

The cases bear only on Flattening's non-crashing content. In the models of the cases every world can access every world, so 4 and H hold, and so do both Crashing principles. This gives an observation.

Observation 2. C2 plus Crashing Cautious Importation and Crashing Cautious Exportation does not prove Flattening.

The frame of the Reunion model validates both Crashing principles but not Flattening (checked over all valuations). This fits Dorr and Mandelkern's corollary that C2.F is not C2 plus modal principles (29). My cases are consistent with counterfactuals obeying C2 together with 4 and H. Whether they do is a separate question. Dorr and Mandelkern note that complex left-nested conditionals are hard to assess (35), in connection with Sequentiality. 4 and H become left-nested once the defined operator is unpacked, so the same caution plausibly applies to them. That is my extension of their remark.

Theorem 1 holds for any conditional that obeys C2 and either half of Flattening, so one may ask about indicatives. The indicative pair, "If at least one of them came, then if Ann came, Bob was there" and "If Ann came, Bob was there", does not split for a speaker who does not know who came. Supposing indicatively that at least one came does not select Bob's coming; it leaves open which of them came. So condition (b) fails. In the Reunion both sentences are unwarranted, and in Avoidance both are acceptable in their negative forms. This gives no evidence against Flattening for indicatives. It gives none for it either.

It also does not answer Dorr and Mandelkern's open problem. Restricting an information state by p and then by q is restricting it by pq, for any q. A semantics built from successive restriction, or from conditioning, therefore validates full Import-Export, which collapses C2 into the material conditional (21). An explanation of Flattening for indicatives must separate the sequence "p, then pq" from "p, then q". I do not supply one. Nor does Stalnaker's Thesis supply one. Dorr and Mandelkern show that under a restricted Thesis the two sides of suitable instances receive equal probability, but they find no route to probability one. Their tree models satisfy a strong form of the Thesis while Flattening instances with Boolean p, q, r have probability below one (44 n. 36).

The consequence for the unity of "if" follows from their own reasoning. They decline to validate Import-Export because it fails for subjunctives, and they explain its indicative appeal by mood. They exempt Flattening because it "appears valid for both" (22). If the argument above is right, Flattening fails for subjunctives too, so the same reasoning applies. A unified "if" should not validate Flattening, and whatever appeal it has for indicatives should be explained by what the indicative contributes. That leaves C2.F's model theory intact as a theory of a conditional. It withdraws the claim that C2.F is part of the logic of "if" in both moods.

8. Conclusion

The matched cases isolate a proposed failure of Flattening under scenario-relative interpretation. Their finite models agree on all first-degree counterfactuals while separating the nested sentence from its flattening partner. The control removes several simple alternative readings; Buttons adds a conditional-selection argument based on local changes; and Theorem 1 displays the associated commitment of either half of Flattening within C2.

The decisive outstanding issue is interpretation. The formal models do not prove that ordinary English expresses their selection functions. Persistent shifts face the stated

discourse test, while temporary contextual shifts and rejection of the proposed nested judgment remain possible. The contribution is therefore a structured challenge and an explicit comparison between predictions, not a completed empirical case against sequence semantics. If the stable scenario-relative reading is sustained, Flattening cannot be a law of that reading of the counterfactual. If it is not, the natural-language objection fails while the mathematical separation remains.

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